Exercise 6 - Topic Modeling

Topics and Trends in NLP

Deadlines

The deadline for Exercise 6 is: **Sunday, 14.12.2025, 23:59 (Zurich Time)**

The deadline for the peer review is: **Monday, 22.12.2025, 23:59 (Zurich Time).**

Note, the peer reviews will be assigned one day after the exercise deadline. You will find instructions for the peer review process at the end of this document.

Learning goals

This exercise is about topic modeling, more specifically about Latent Dirichlet Allocation (LDA) and topic modeling based on pre-trained language models (PLM). By completing this exercise, you should…

- understand how topic modeling is used as a text-mining tool.
- be able to apply LDA and PLM-based topic models.

Please keep in mind that you can always consult and use the exercise forum if you get stuck (note that we have a separate forum for the exercises).

Deliverables

Your solutions for each part should be submitted as self-contained iPython notebooks (ipynb files). The notebooks should contain your well-documented, EXECUTED, and EXECUTABLE code. That way, your reviewers can view and execute your code without potential dependency issues and/or installing new packages or versions of packages. We encourage you to solve the exercise on Google Colab or Kaggle and download the .ipynb file when everything is completed.

You will also have to write a short lab report at the end of each script. The lab report should be written in a markdown cell and contain a detailed description of the approaches you used to solve this exercise. You can discuss what worked and what didn't work, but in both cases, you please provide reasoning as to why something did or didn't work. Please also include the results. Inside each notebook, we have provided specific questions denoted with "📝❓" that should be addressed in the report.

Please hand in your code in a notebook and name it as follows. The notebook should contain a lab report at the end in a markdown cell.

- **ex06_tm.ipynb**

Data

For this exercise, you will work with the dblp: a large database containing metadata on computer science (and related) publications. You will perform topic modeling on the titles (not on the full text itself!) of computer science publications to detect important topics and trends in the field.

Given

For the exercise, you are given this notebook. It already contains code for downloading the dataset, and instructions on how to proceed.

Part 1: Topic Modeling using LDA

The given notebook limits the number of titles to a reasonable amount and divides publications into three time periods: before 1990, from 1990 to 2009, and from 2010 onwards. Perform topic modeling on the three time periods.

Please experiment with varying numbers of topics (starting with at least five) and with different ways of preprocessing.

For each period, assign a name to each generated topic based on the topic's top words. List all topic names in your report. If a topic is incoherent to the degree that no common theme is detectable, you can just mark it as incoherent (i.e., no need to name a topic that does not exist).

- Do the topics make sense to you? Are they coherent? Do you observe trends across different time periods? Discuss in 4-6 sentences.

Part 2: Topic Modeling using Combined Topic Models (CTMs)

Bianchi et al. 2021 propose a topic modeling approach that makes use of pre-trained language models such as BERT. The authors provide a simple colab tutorial demonstrating how to use the CTM library that implements their method.

Again, perform topic modeling for the 3 time periods. This time using the CTMs. Use the same number of topics as before. You can copy and adjust code from the author's tutorial. We suggest using "sentence-transformers/paraphrase-mpnet-base-v2" for CTM.

- Again: Assign a name to each topic based on the topic's top words (for each period). List all topic names in your report.
- Bianchi et al. 2021 claim that their approach produces more coherent topics than previous methods. Let's test this claim by comparing the coherence of the topics produced by CTM with the topics produced by LDA. Describe your observations in 3-4 sentences.
- Do the two models generate similar topics? Can you discover the same temporal trends (if there are any)? Discuss in 5-6 sentences.
- Can you suggest an alternate model apart from paraphrase-mpnet-base-v2? What could be some of the possible advantages and disadvantages of using the chosen model? Hint: Look at some of the models here. Note: You do not need to execute the code for an alternate model.

Submission & Peer Review Guidelines:

For all exercises, peer reviews will be carried out on OLAT.

On the day after the deadline for handing in the exercise expires at 8 PM, you will have time to review the submissions of your peers. You need to do **1 review per person** to get the maximum points for this exercise.

Here are some additional rules:

- **All file submissions are anonymous (for peer review purposes): Do not write your name into your Jupyter Notebooks.**
- **EVERY** member of each team submits on OLAT.
- Please submit a zip folder containing all the deliverables.

Groups & Peer Reviews:

- You should work in the same group for all exercises. Each member should contribute equally!
- If you do not submit your peer review, you will receive a -0.25 point penalty on the final grade of the exercise.
- Please use full sentences when giving feedback. Please write the peer review in English.
- Provide **constructive** and helpful comments to your peers! If you criticise something, you should also be able to provide actionable and realistic suggestions for how

something can be improved.
- Please answer all the pre-designed review questions of the peer review to the best of your ability.

Declaration of Usage of Generative AI

For every exercise, please append a short **“Declaration of Usage of Generative AI”** section at the end of
your submission. In this declaration, briefly (a few sentences are sufficient!) describe **how** and to **what extent**
you used Generative AI tools (e.g., ChatGPT, Copilot, Gemini, etc.) during the completion of this exercise.

Your statement should address the following points:

- **Source code and documentation:** How was Generative AI used to assist in developing, debugging, or documenting your code?
- **Written report:** Was Generative AI used to draft, revise, or post-edit the written components of your submission?
- **Learning support:** Did you use Generative AI as an assistant for concept clarification, idea exploration, or learning guidance?